Internship Overview

Autonomous Field Navigation of Robot

[ID 022026]

Develop an autonomous navigation stack for a small field robot, using LiDAR and positioning sensors for obstacle detection and safe movement.

Objectives

- Enable autonomous field navigation
- Detect obstacles using LiDAR data
- Fuse localization and perception signals for safer motion

Tasks

- Integrate LiDAR, GPS, and Pixhawk hardware interfaces and sensor data
- Build localization and navigation modules
- Build obstacl detection module from LiDAR observations
- Deploy autonomous path module
- Calibrate sensors, Validate navigation performance on the robot chassis

Requirements

- Python programming
- Robotics fundamentals
- ROS basics
- Path planning fundamentals

Equipment

- RPLIDAR C1 LiDAR
- ZED-F9P-01B-01 GPS module
- Holybro Pixhawk 6X flight controller
- UGV Rover ROS 2 Open-source 6 Wheels 4WD

Deliverables

- Autonomous navigation prototype with LiDAR-based obstacle detection
- Field validation results

Work Breakdown

- 30% hardware integration
- 70% coding

Duration: 2 months

Payment: Available in Bizquick

Supervisor: Kaleb Granados